

SAMPLE PAPER-I

- Please check that this question paper contains **4** printed pages.
- Check that this question paper contains **19** questions.
- Write down the Serial Number of the question in the left side of the margin before attempting it.

CLASS-IX SUBJECT: MATHEMATICS

Time Allowed:1 Hr 30 Mins

Maximum Marks: 40

General Instructions:

1. This Question Paper has 5 Sections A- E.
2. Section A has 10 MCQs carrying 1 mark each.
3. Section B has 2 questions carrying 2 marks each.
4. Section C has 4 questions carrying 3 marks each.
5. Section D has 2 questions carrying 5 marks each.
6. Section E has 1 case based integrated units of assessment of 4 marks with sub- parts of the values of 1, 1 and 2 marks respectively.
7. All Questions are compulsory. However, an internal choice in 1 Question of 5 marks, 1 Question of 3 marks and 1 Question of 2 marks has been provided.
An internal choice has been provided in the 2marks questions of Section E.
8. Draw neat figures wherever required.

SECTION - A (1M×10 = 10M)

1. The point A(5,3) is reflected first in the x axis and then in the Y axis. What are the final coordinates?

A(5,3). B (-5,-3) C (-5,3) D (3,-5)

2. The vertex of a rectangle are (2,1),(2,6) and (8,1). The fourth vertex is A(6,8) B(8,6).
C(1,8) D(6,2)

3. Which of the expression is not a polynomial?

A. $4x^3 - 2x + 1$

B. $x^2 + \frac{1}{x}$

C. $7x - 9$

D. 6

4. If $P(x) = x^2 - 9$ then $P(3)$ is equal to

A. 18 B. 9 C. 0 D. -9

5. A polynomial $P(x)$ satisfies $P(2) = 0$, which of the following statements is true

A $x=2$ is zero of the polynomial.

B The constant term is 2

C At $x=2$ the value of the polynomial is 0

D The degree of polynomial is 0

6. Decimal representation of rational numbers cannot be

A) Terminating

B) Non-Terminating

C) Non-Terminating Repeating

D) Non-Terminating Non-Repeating

7. Which number lies between 3 and 4

A $\sqrt{7}$

B. $\sqrt{8}$

C. $\sqrt{10}$

D $\sqrt{15}$

8. If $a+b = 12$, $ab = 20$, then a^2+b^2 is

A. 104

B. 124

C. 144

D 164

9. Which of the following expression is always a perfect square?

A. x^2+y^2

B. $x^2+2xy+y^2$

C. $x^2 + xy + y^2$

D. $x^2 - y^2$

10. **Directions:** In the following question, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Assertion: The point (3,5) and (-2,5) are at equal distance from x axis.

Reason: The distance of a point from x axis depends only on the absolute value of its Y coordinate.

SECTION - B (2M×2 = 4M)

11. Two students draw the graph of the polynomial $P(x) = 4x+1$. They choose the points (1,5) and (3,13).

(a) Verify that both points satisfy the polynomial.

(b) Find slope of the graph.

12. Find the possible expression for length and breadth of the rectangle whose area in square unit is

$$25a^2 - 30ab + 9b^2$$

OR

Simplify $\frac{4x^2+4x+1}{4x^2-1}$

SECTION - C (3M×4=12M)

13. Plot the point A(2,1) B(-1,2), C(-2,-1), and D(1,-2) in the coordinate plane. Is ABCD a square? Can you explain why? What is the area of this square?

14. If $a + b + c = 5$, $ab + bc + ca = 10$, then find the value of $a^3 + b^3 + c^3 - 3abc$.

15. Prove that $\sqrt{3}$ is an irrational number.

OR

15b i) Convert it in the form of p/q $23.\overline{43}$

ii) Find the decimal value of $1/7$ by long division, then find the value of $2/7, 3/7, 4/7, \dots, 6/7$ Without dividing, write your observation.

16. If the temperature of a liquid can be measured in kelvin units x k and in Fahrenheit unit as y° f, the relation between the two systems of measurement of temperature is given by the linear equation $y = \frac{9}{5}(x - 273) + 32$.

- i) Find temperature of the liquid in Fahrenheit if the temperature of liquid is 313 k.
- ii) If the temperature 158° f, find the temperature in kelvin

SECTION - D (5M × 2 = 10M)

17. a) A square playground has a side of $(x-6)$ m. After renovation the side becomes $(x+6)$ m. Using the above data.

- i) Find the increase in the area using an algebraic identity.
- ii) A square hall has a side length of $(2x+3)$ m. The cost flooring is ₹100 per square metre. Find the cost of flooring when $x = 4$.

OR

17. b) i) Factorise the given expression: $x^2 + \frac{5}{6}x + \frac{1}{6}$

ii) Find possible expressions for the length and breadth of the rectangle whose area is given by $25a^2 - 30ab + 9b^2$

- 18. i) Find two rational numbers and irrational numbers between 4 and 5.
- ii) Construct square root of 2 units on a number line.

SECTION - E **(1M+1M+2M = 4M)**

19. A school has planned a rectangular garden on a coordinate grid. The four co-ordinates are A(1,2) B(7,2) C(7,6) and D(1,6)

- i) In which quadrant does the garden lie?
- ii) Find the length and breadth of the garden.
- iii) Find coordinates of diagonal AC

Or

Calculate the area and perimeter of the garden.

----- XXX -----
